

HAT-P-19b

R-1276

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-19_231013 · created 2026-07-31T14:29:05+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460230.7943 +/- 0.0048 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.007 +/- 0.053
Transit depth (Rp/Rs)^2	0.0054 +/- 0.077 [%]
Orbital Inclination (inc)	85.47 +/- 1.65
Airmass coefficient 1 (a1)	0.67 +/- 0.29
Airmass coefficient 2 (a2)	0.33 +/- 0.92
Scatter in the residuals of the lightcurve fit is	41.31 %
Best Comparison Star	None
Optimal Aperture	1.46
Optimal Annulus	4.5
Transit Duration (day)	0.035 +/- 0.039

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.007 ± 0.053 vs 0.1418 (deviation 2.54σ)
Duration fit vs published	0.035 d vs 0.1196 d (deviation 2.17σ)
Mid-time O–C	-1.4 min
Depth SNR	0.1
Residual scatter	41.31 %

Light curve

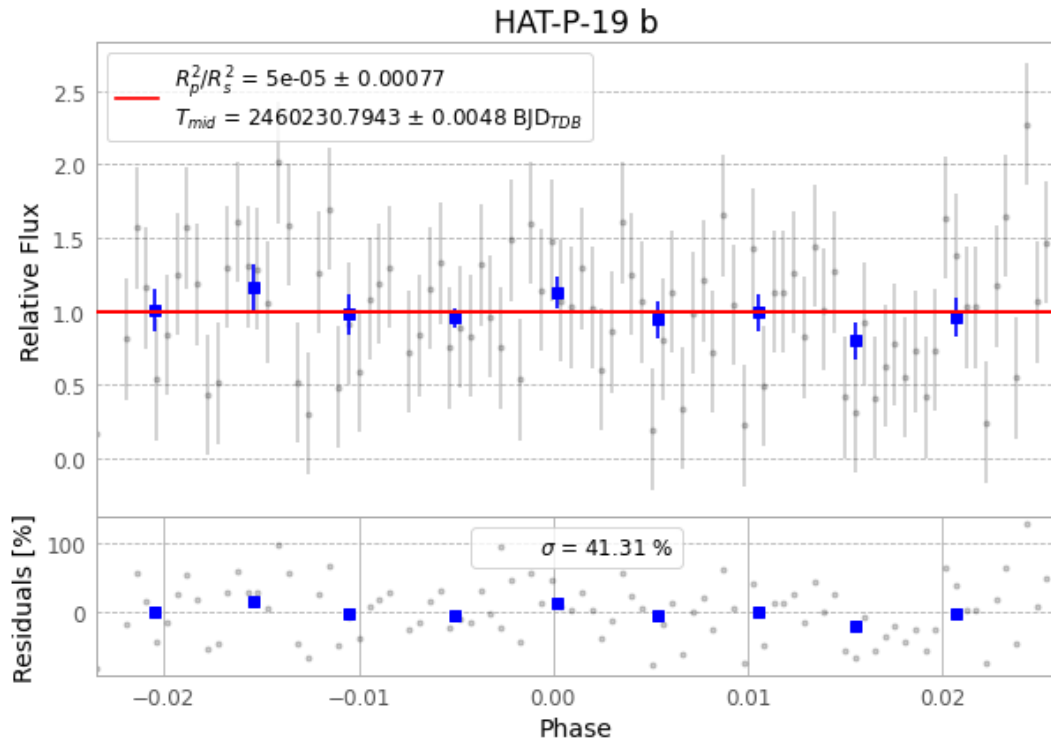


Figure 1 — FinalLightCurve_HAT-P-19 b_2023-10-12.png (SHA-256 7852c4fd4f73adee...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [337, 338], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 912ff7556c8eb2da...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

96 FITS frames (63 MB), HATP-19231013043916.FITS → HATP-19231013092415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-19-231013.log (9a9be655a5d1...), FinalLightCurve_HAT-P-19 b_2023-10-12.png (7852c4fd4f73...), FinalLightCurve_HAT-P-19 b_2023-10-12.csv (aa0782364d67...)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 14:29Z.